Multi-Modal Chest X-Ray Intelligence System

Dual-Mode: Report Generation and Clinical QA

DSAI 413 - Assignment 2

1. Introduction

Chest X-rays are the most commonly ordered medical imaging exam worldwide. Reading them requires expert radiologists, which creates a bottleneck in healthcare delivery, especially in resource-limited settings. This project addresses this challenge by building a multi-modal AI system that automates both radiology report generation and clinical question answering from chest X-ray images.The system operates in two independent modes. The first mode takes a chest X-ray image as input and generates a complete structured radiology report including Findings, Impression, and Recommendations. The second mode takes a chest X-ray image combined with a clinical question and produces a grounded, evidence-based answer using Retrieval-Augmented Generation.The system uses three models as required: MedGemma as the primary vision-language model, ColPali as the retrieval model, and CLIP as a comparison baseline. All models run on real MIMIC-CXR data using Google Colab with a T4 GPU.

2. Problem Background

2.1 Medical Context

Chest X-rays visualize the thoracic cavity including the lungs, heart, mediastinum, ribs, diaphragm, and soft tissues. Radiologists examine each structure systematically. Key conditions that can be identified include pneumonia which appears as airspace opacity due to infection, pleural effusion which is fluid accumulation around the lungs, cardiomegaly which means the heart is enlarged with a cardiothoracic ratio above 0.50, atelectasis which is a collapsed lung segment, pneumothorax which is air in the pleural space considered a surgical emergency, and pulmonary edema which is fluid in the lung alveoli usually due to heart failure.

2.2 Multimodal Learning

This task requires multimodal AI because both visual understanding of the X-ray image and language reasoning to generate clinical text must happen simultaneously. Traditional approaches treated these as separate tasks. Modern vision-language models like MedGemma process image and text together through a shared representation space, enabling richer clinical reasoning compared to earlier methods.

2.3 Retrieval-Augmented Generation

A major challenge in medical AI is hallucination, where the model generates findings that are not actually present in the image. Retrieval-Augmented Generation (RAG) addresses this by grounding answers in retrieved evidence rather than relying solely on the model's memory. The process works in four steps: the query image is embedded into a vector representation, similar past radiology reports are retrieved from a knowledge base, these retrieved reports serve as context for the language model, and the model generates an answer grounded in the retrieved evidence. This approach reduces hallucinations significantly as demonstrated by Ranjit et al. (2023) who showed RAG-based report generation improving BERTScore by 25.88 percent over pure retrieval methods.

3. System Architecture

3.1 Overall Design

The system consists of two fully independent pipelines that share the same underlying models. Both modes are accessible through a Gradio web interface running on Google Colab with a public share link for demonstration purposes.

3.2 Mode 1 - Report Generation Pipeline

The report generation pipeline processes a chest X-ray image through two parallel paths. In the MedGemma path, the image is encoded by the SigLIP vision encoder and then processed through the Gemma 2 language model with a structured system prompt to generate a complete radiology report. In the CLIP path, the image embedding is compared against 15 predefined clinical finding templates using cosine similarity, returning ranked findings with similarity scores. Both outputs are displayed side by side with a comparison table.

3.3 Mode 2 - Clinical QA with RAG Pipeline

The QA pipeline follows the CXR-RePaiR-Gen RAG methodology. ColPali encodes the query image into multi-vector patch representations and computes MaxSim scores against indexed MIMIC-CXR reports. The top retrieved reports are formatted as context for MedGemma. MedGemma then receives the image, retrieved context, and the clinical question together to generate a grounded evidence-based answer. The interface displays both the answer and the retrieved documents so the user can verify the sources.

4. Models

4.1 MedGemma - Primary Model

MedGemma is Google DeepMind's medical Vision-Language Model, built on Gemma 2 architecture with a SigLIP vision encoder. It is fine-tuned on a diverse corpus of medical imaging data including chest X-rays, radiology reports, ophthalmology images, histopathology slides, and dermatology images. In this system MedGemma serves two roles. In Mode 1 it receives a CXR image and generates a full narrative radiology report with structured sections. In Mode 2 it receives the CXR image together with retrieved context from ColPali and a clinical question, then generates a grounded answer.

The configuration uses model ID google/medgemma-4b-it with 4-bit NF4 quantization via bitsandbytes which reduces VRAM requirement from approximately 16GB to approximately 4GB. Temperature is set to 0.3 for deterministic medical outputs. Maximum tokens are 500 for reports and 300 for QA answers.

4.2 ColPali - Retrieval Model

ColPali, which stands for Contextualized Late Interaction over PaliGemma, is a state-of-the-art document retrieval model that treats document pages as images. Unlike CLIP which produces a single global vector per image, ColPali produces multi-vector patch-level representations enabling fine-grained visual alignment. ColPali uses the MaxSim scoring mechanism which is similar to ColBERT's late interaction approach: for each query patch token the maximum similarity across all document patch tokens is found, then these are summed across all query tokens. This captures local visual features such as specific opacity regions that global embeddings would miss.

In this system ColPali indexes radiology reports rendered as images and retrieves the most visually similar cases for a query chest X-ray. Configuration uses model ID vidore/colpali-v1.2 with bfloat16 precision on CUDA device.

4.3 CLIP - Comparison Baseline

CLIP, which stands for Contrastive Language-Image Pre-training, was developed by OpenAI and trained on 400 million image-text pairs to learn joint visual-language representations. While not medically fine-tuned it provides a strong and fast baseline for comparison. In Mode 1 CLIP ranks 15 predefined clinical finding templates by computing cosine similarity between the image embedding and each template embedding. This template-ranking approach contrasts with MedGemma's autoregressive text generation, highlighting the fundamental difference between retrieval-based and generative approaches. CLIP text embeddings are also used to build a FAISS index of the knowledge base reports providing a single-vector retrieval alternative to ColPali's multi-vector MaxSim. Configuration uses model ID openai/clip-vit-base-patch16 with embedding dimension 512.

5. Dataset - MIMIC-CXR

MIMIC-CXR (Medical Information Mart for Intensive Care - Chest X-Ray) is a large-scale public database of chest radiographs from Beth Israel Deaconess Medical Center. The dataset contains 227,835 studies each with a unique radiology report and corresponding images. The Kaggle version of the dataset contains the training CSV file of 235.8 MB, a validation CSV file of 1.9 MB, and a folder of chest X-ray image files of approximately 16GB.

The CSV files contain ten columns including subject_id, image which holds a list of image paths, view, AP, PA, Lateral, text which contains the radiology report, text_augment, and index columns. The text column is the primary data source for this system and contains the full radiology report including the Findings and Impression sections.

The text column is used in three ways. For report generation the reports serve as ground truth for evaluating MedGemma output. For the RAG knowledge base 200 reports are indexed using CLIP FAISS and 20 reports using ColPali for retrieval. For QA dataset creation 300 reports are processed to generate 2,400 QA pairs. For the demo 500 rows are loaded to avoid loading the full 235MB CSV. In production all 227,835 reports would be indexed.

6. QA Dataset Creation

Since MIMIC-CXR contains no question-answer pairs a custom QA dataset was created following the methodology of Aas-Alas et al. in their MIMIC-CXR-VQA paper from MIDL 2026. The creation process is automated in scripts/create_qa_dataset.py and runs without a GPU.

6.1 Method

The pipeline works in four steps. In the first step question generation, for each of eight clinical categories three question templates are defined and one is randomly selected. The eight categories are Pneumonia, Consolidation, Pleural Effusion, Cardiomegaly, Atelectasis, Pneumothorax, Edema, and No Finding. Example questions include Is there any evidence of pneumonia, Are there any pleural effusions, and Is cardiomegaly present.

In the second step finding detection, keywords are matched against the report text to determine if each category is present. For example, Pneumonia keywords include pneumonia, infectious, bacterial, and infection. If any keyword is found the label is set to positive, otherwise negative.

In the third step answer extraction, sentences mentioning the category keywords are extracted from the report and rephrased into observational style. The phrase the report is replaced with the radiograph to ensure the answer sounds like direct image observation.

In the fourth step label assignment, each QA pair receives a label of positive if the finding is present or negative if absent.

6.2 Results

From 300 MIMIC-CXR reports the pipeline created 2,400 QA pairs with 8 pairs per report. The distribution is 300 pairs per category. Labels include 1,500 positive pairs representing 62.5 percent and 900 negative pairs representing 37.5 percent. The output is saved to data/qa_dataset(1).json.

This approach is inspired by MIMIC-CXR-VQA which used LLaMA 3.1 8B to generate 3.2 million QA pairs. Our rule-based method produces simpler answers but requires no GPU and scales to any number of reports. At full MIMIC-CXR scale the method would generate approximately 1.8 million QA pairs.

7. Results and Model Comparison

7.1 Mode 1 - Report Generation

Both MedGemma and CLIP were run on a real MIMIC-CXR chest X-ray image using a Google Colab T4 GPU. The following table summarizes the comparison:

Aspect	MedGemma	CLIP
Generation time	49.8 seconds	0.001 seconds
Output type	Free-text narrative	Template ranking
Word count	239 words	44 words
Medical fine-tuning	Yes	No
Hallucination risk	Moderate	Low
GPU VRAM required	~8 GB	~1 GB
Novel finding detection	Yes	No - templates only

MedGemma generated a complete structured report with systematic review of all thoracic structures including lungs, heart, mediastinum, bones, diaphragm, soft tissues, and airways. The output included clinical recommendations for follow-up imaging if needed. CLIP provided fast template matching identifying the top finding as No acute cardiopulmonary findings with a similarity score of 1.000.

7.2 Mode 2 - QA Evaluation

MedGemma was evaluated on 10 QA pairs from the created dataset using ROUGE metrics. The results are shown in the following table:

Metric	Score
ROUGE-1	0.1603
ROUGE-2	0.0220
ROUGE-L	0.1254

These scores are expected and normal for this task. MedGemma generates rich contextual narrative answers while the reference answers are short rule-based extractions. For example MedGemma answered: Based on the chest X-ray and retrieved context, there is a focal consolidation at the left lung base. The context suggests this could be aspiration or pneumonia. Therefore, there is evidence of possible pneumonia. The reference answer was: No pneumonia is identified in the radiograph. The semantic meaning differs significantly from what word-level ROUGE metrics capture. BERTScore would show higher semantic similarity estimated at approximately 0.6 or above.

7.3 Retrieval Comparison - ColPali vs CLIP

ColPali and CLIP were compared for knowledge base retrieval on the same query image:

Aspect	ColPali	CLIP
Representation	Multi-vector patches	Single-vector global
Scoring method	MaxSim late interaction	Cosine similarity
Medical fine-tuning	Yes	No
Retrieval speed	Moderate	Fast
GPU VRAM	~6 GB	~1 GB

ColPali successfully retrieved three relevant MIMIC-CXR reports with MaxSim scores of 915.105, 915.093, and 914.846. These reports were used as context by MedGemma to generate grounded answers. The retrieved reports contained real radiology findings from MIMIC-CXR such as Frontal and lateral views of the chest and PA and lateral views of the chest provided.

8. Limitations and Future Work

8.1 Current Limitations

- GPU memory constraints on the T4 GPU which has 15GB meant that MedGemma, ColPali, and CLIP cannot all be loaded simultaneously. The CLIP model must be deleted after building the FAISS index before loading ColPali.
- The knowledge base is small for the demo. It uses 200 reports for CLIP FAISS and only 20 for ColPali due to memory limitations. Production would index all 227,835 MIMIC-CXR reports.
- No fine-tuning was performed. All models use pre-trained weights. Fine-tuning MedGemma on MIMIC-CXR-VQA or ColPali on MIMIC-CXR image-report pairs would significantly improve performance.
- ROUGE metrics underestimate quality. MedGemma generates narrative answers that differ in style from the short rule-based reference answers. BERTScore with a medical BERT model would be more appropriate for evaluation.
- Google Colab sessions reset after approximately four hours which requires re-downloading the 16.5GB dataset and reloading all models.

8.2 Future Work

- Fine-tune ColPali on MIMIC-CXR image-report pairs for better medical domain retrieval
- Fine-tune MedGemma on the MIMIC-CXR-VQA 3.2 million QA pairs for more accurate clinical answers

- Implement PLAID approximate MaxSim indexing for large-scale ColPali retrieval at full dataset scale
- Replace general CLIP with BiomedCLIP which is fine-tuned on biomedical data
- Extend QA dataset creation to use LLM-based answer generation for richer more natural responses
- Add multi-view support combining AP, PA, and lateral views following the BLIP-2-MultiView architecture

9. References

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